Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/29/2008
PI Name:	Minna, John D. M.D.		
Project Title:	NSCOR: Lung Cancer Pathogenesis and HZE Particle Exposure		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Dallas	**State:**	TX
Zip Code:	75390-8593	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2004 NSCOR Space Radiation NNH04ZUU002N
Start Date:	03/31/2005	**End Date:**	03/31/2010
No. of Post Docs:	9	**No. of PhD Degrees:**	1
No. of PhD Candidates:	7	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Start/end dates changed to 3/31/2005-3/31/2010 per grant documents from PI (12/06)		
Key Personnel Changes/Previous PI:	We have made several administrative changes during the past reporting period. In 2007 Yang Xie, Ph.D. joined the Biostatistical Bioinformatics Core D and provides special expertise in analysis of array expression profiling data. Cheng-Ring Yang, Ph.D. joined the Genomics and Proteomics Core (Core C) to develop and provide expertise on proteomics profiling. Mitsuo Sato, M.D., Ph.D. returned to a faculty position in Japan and his work in Projects 1 and 3 are being taken over by Asst. Professor of Pharmacology Boning Gao, Ph.D. who is also expert in the HBEC system that Dr. Sato had provided.		
COI Name (Institution):	Burma, Sandeep (The University of Texas Southwestern Medical Center) Chen, David (The University of Texas Southwestern Medical Center) Gazdar, Adi (The University of Texas Southwestern Medical Center) Girard, Luc (The University of Texas Southwestern Medical Center) Richardson, James (The University of Texas Southwestern Medical Center) Shay, Jerry (The University of Texas Southwestern Medical Center) Story, Michael (The University of Texas Southwestern Medical Center) Wright, Woody (The University of Texas Southwestern Medical Center) Xie, Xian-Jin (The University of Texas Southwestern Medical Center) Minna, John (The University of Texas Southwestern Medical Center)		

	Yang, Chin-Reng (The University of Texas Southwestern Medical Center)
Grant/Contract No.:	NNJ05HD36G
Performance Goal No.:	
Performance Goal Text:	
Task Description:	The University of Texas Southwestern Medical Center (UTSW) NSCOR focuses on the development of quantitative dose risk estimates following HZE particle irradiation for the development of key genetic, epigenetic, gene expression, and cellular functional changes in the multi-step pathogenesis of lung cancer in both new human bronchial epithelial cell (HBEC) and transgenic mouse models of lung cancer. The risk of developing these changes will also be compared in these models to that of gamma-irradiation. These dose risk assessments will measure events of both radiation-induced promotion (modification of proliferation kinetics of already-initiated cells) as well as radiation-induced initiation (mutational) events. They will also measure effects on development of cancer "stem-like" cells ("cancer initiating cells"). 3D organotypic culture and animal models will also allow measurement and risk estimation in both tissue surrogates and in vivo tissue situation including measurement of "bystander" effects. These individual risk estimates can then be combined into a model (such as a two-stage clonal expansion model) for overall risk of developing lung cancer from exposure to galactic cosmic radiation. To achieve these goals, UTSW has assembled a team of scientists who are leaders in the study and translation application of the molecular pathogenesis of lung cancer (Minna, Gazdar, Shay); in radiobiology and DNA repair (Chen, Story); a panel of expert Internal and External Advisors (from other NSCORS and lung cancer research experts); as well as expert consultants and collaborators. This team has developed a novel immortalized HBEC system that can be genetically manipulated, studied in 2D monolayer and 3D organotypic cultures. Using these HBECs we have preliminary studies with HZE particle irradiation showing that further progression toward lung cancer can be detected and quantified, that specific expression profiles for HZE irradiation exist, and that HBECs genetically manipulated to progress part of the way toward malignancy are even more sensitive to HZE-induced functional changes. In addition, we have initiated experiments testing whether several transgenic mouse models of lung cancer after low and high LET exposure increase the early onset of lung cancer in the whole animal in real time. The NSCOR has 4 Projects: 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation; 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells; 3. Effect of HZE particle irradiation on functional progression of human lung cancer at the cellular and organotypic level; and 4. Effects of HZE particles on the development of lung cancer in vivo in novel mouse models. These projects are supported by 4 Cores: Administrative; Cell Culture; Genomics and Proteomics; and Biostatistics and Bioinformatics. In addition, with funding from the Department of Energy (DOE) contributing to this NSCOR, studies of low-LET and low dose rate radiation in the same model systems are also being undertaken. The HZE particle irradiation will be done at Brookhaven National Lab, and the biostatistical analysis shows the experiments are powered to help achieve NASA mandated risk estimate confidence levels. Finally, since this group has substantial UTSW institutional commitments, holds a NCI Special Program of Research Excellence (SPORE) in Lung Cancer, a Center for the NCI Early Detection Research Network (EDRN) program, and is part of the NCI sponsored Genetic Epidemiology of Lung Cancer Consortium (GELCC) there will be great synergism and additional resources available for the successful completion of this proposal.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Lung cancer is the leading cause of cancer death in both women and men in the USA and in the Western world. While lung cancer is caused by smoking exposure in 85% of people, because of the large numbers the remaining 15% of cases arising in life time never smokers (~25,000 patients per year) is a sizable health problem in the USA. Likewise, 50% of all new lung cancer cases occur in former smokers (quitting over 5 years before). In all of these scenarios a major underlying question has been the role of other environmental carcinogens especially environmental radiation. One source of this is radon gas in the environment including radon gas exposure in homes with alpha particles (which represent high LET radiation). Thus, the interaction of radiation in the environment on Earth including from alpha particles in the genesis of lung cancer and quantitation of the effect of such radiation on lung epithelial cells is of major health importance. Radiation is a known carcinogenic influence but the molecular events in lung epithelial cells in response to radiation need to be determined. A key element of this is the impact of radiation on lung epithelial cells with preneoplastic lesions. Our NSCOR uses innovative technology of human bronchial epithelial cells (HBECs) developed by us (coming from 45 different individuals) which have also been engineered to have known oncogenic preneoplastic changes (such as oncogenic KRAS, knockdown of or expression of mutant p53, expression of oncogenic mutant EGFR, expression of c-myc). Our studies are providing quantitative data following high LET irradiation on these HBECs. The primary benefits from this work will be the development of quantitative risk assessment models for astronauts of high LET radiation in space. However, the results will also have major impact and benefit to life on Earth includes new knowledge of the effects on human lung epithelial cells of irradiation in terms of quantitative genetic and epigenetic changes and gene expression changes following irradiation which are of potential importance as markers of radiation exposure that could occur through environmental exposures, accidentally or through terrorism. As part of this will be the quantitative response of DNA repair pathways to radiation including specific DNA repair components such as DNA repair enzymes and signaling pathways in lung epithelial cells. We are also determining quantitative changes in biologic phenotypes of lung epithelial cells in response to radiation which include colony formation in liquid and soft agar (anchorage independent growth) and differentiation and invasion in 3 dimensional cultures, and finally growth in vivo as either xenografts (for human cells) or endogenous lung cancers (for mouse models of lung cancer). From all of the above information we will identify novel molecular biomarkers (mRNA, DNA, protein) for lung carcinogenesis. Finally, we are also learning about inter individual variation of the response of HBECs to radiation with respect to the various genetic, epigenetic, mRNA and protein expression, and signaling pathway changes. Our large HBEC panel coupled with new high density single nucleotide polymorphism (SNP) analysis will allow describing polymorphic differences in such responses.

Task Progress:

Project 1. Genetic and epigenetic changes in human bronchial epithelial cells following exposure to HZE particle irradiation.
Specific Aim 1: To quantitate the effects of HZE particle irradiation in causing genetic changes in HBECs and their isogenic derivates.

Specific Aim 2. To quantitate the amount of epigenetic changes caused by HZE particle irradiation in HBECs and their isogenic derivates using high throughput quantitative methylation specific PCR (MSP) assays for acquired promoter methylation

Specific Aim 3. To quantitate the effects of HZE particle irradiation in causing expression profile (mRNA and protein) changes in human bronchial epithelial cells (HBECs) and their isogenic derivates containing various lung cancer-related mutations using array based genome wide and multi protein detecting approaches.

Specific Aim 4. To determine the effect of inter-individual variation on the effects of HZE particle irradiation in causing genetic changes, epigenetic changes, mRNA and protein expression profile changes in human bronchial epithelial cells (HBECs).

We continued studies of gene expression changes following various types of radiation outlined in Aim 3. mRNA gene expression profiles were studied with Illumina v2 arrays before and at different time points after radiation. Supervised clustering methods identified samples by the radiation received via their gene response over time. We also examined kinetic differences rather than static differences as described above. The overall pattern of expression of genes was used to develop predictors of radiation response. From 1480 genes, temporal patterns were developed and 322 identified 12 different patterns

Project 2. Effect of HZE particles on DNA damage-sensing and repair pathways in human lung epithelial and fibroblast cells.

Aim 1. To test the hypothesis that the immediate cellular response to DNA damage induced by HZE would be different and more complex than that induced by X-rays.

Aim 2. To test the hypothesis that persistent DNA damage induced by HZE is more complex and difficult to repair than that induced by X-rays and, therefore, more deleterious.

Aim 3. To determine if damage induced by HZE is preferentially repaired by non homologous end joining (NHEJ) or homologous recombination (HR) or is preferentially repaired in certain phases of the cell cycle.

Aim 4. To compare the response of human bronchial epithelial cells as non immortalized, immortalized, and with premalignant oncogenic changes in 3D cultures to HZE exposure using a high throughput approach.

Both X-rays and 1 GeV Fe particles elicit similar responses at the sites of DNA breaks. However, Fe-induced damage is repaired more slowly and a large fraction of breaks are not rejoined, thereby resulting in the higher relative biologic effect (RBE) of HZE particles. Using charged particles of increasing molecular weights (O vs. Si vs. Fe) we find that the incidence of un-rejoined breaks increases with increasing particle molecular weight presumably due to increasing complexity of the damage induced. We find that the capacity of both lung fibroblasts and HBECS to repair HZE-induced DNA damage is severely limited compared to gamma rays No major differences are seen between 2D and 3D cultures for fibroblasts. The capacity of HBECs to repair gamma-ray induced DNA damage varies depending upon the cancer-promoting oncogenic changes they harbor – these differences are minimized though in response to 1 GeV Fe.

Project 3. Effect of HZE Particle Irradiation on Functional Progression of Human Lung Cancer

Specific Aim 1. Effects of HZE particles on early events in lung cancer progression (soft agar colony formation and other markers of cellular transformation).

Specific Aim 2. Effects of HZE particles on intermediate events in lung cancer progression (invasion in 3 D organotypic cultures).

Specific Aim 3. Effects of HZE particles on late events in lung cancer progression (ability to form tumors in immune deprived mice).

Soft agar colony forming ability increases for both KRAS+p53 lines as a function of time and dose, and that their response is much greater than for HBEC3-KT (without oncogenic preneoplastic changes). There is no radiation-induced increase in soft agar colony forming ability in cells that over-express either wild type or mutant EGFR. Using quantiative methylation specific PCR (MSP) four of eleven markers found to be methylated are RASSF1A, APC, DcR1, and HCAD with a trend of methylation increasing with time after radiation treatment. We are testing the HZE particle and low LET irradiated HBEC cells of various genotypes for their ability to form tumors in immunodeprived mice as a test of the last events to give full tumor progression. We are also determining the effect of irradiation on the number of cancer stem cell like cells (cancer forming cells) using new sphere forming assays and quantitative mRNA expression profiling for a panel of ~30 stem cell genes.

Project 4. Effects of HZE Particles on the Development of Lung Cancer in Novel Mouse Models

Specific Aim 1. Use K-ras transgenic mouse model of human lung cancer to determine effects of low- and high-LET radiation in both living animals and in isolated tissues.

Throughout the second year of the UTSW NSCOR, experimental groups of approximately one hundred LA1 K-ras animals each were established and irradiated. Two groups were shipped to Brookhaven National Laboratory (BNL) on Long Island, NY for the NSRL-06A and NSRL-06C 56Fe high-LET radiation runs and respectively irradiated with a fractionated or single dose of 1.0 Gy 56Fe particles. Another two experimental groups were irradiated at UTSW with either a fractionated or single dose of 1.0 Gy X-rays for analysis of the effects of low-LET radiation. A fifth group of LA1 K-ras animals was left unirradiated for comparison to all irradiation groups. Irradiation of wild type littermates with a fractionated dose but not a single dose of 1.0 Gy 56Fe particles significantly decreases their survival compared to unirradiated wild-type animals. Mutant animals irradiated with either a fractionated or single dose of 1.0 Gy 56Fe particles have significantly decreased lifespan compared to unirradiated mutant LA1 K-ras mice. In comparison, irradiation with 1.0 Gy X-rays does not affect the survival of mutant or wild type littermate LA1 K-ras mice regardless of administration method. Radiation does not appear to affect overall tumor incidence in irradiated mutant LA1 K-ras

mice compared to unirradiated mutant controls. There is a dramatic progression in the grade of lesions compared to unirradiated mutant mice, X-ray irradiated mutant mice, or mutants irradiated with a single 1.0 Gy dose of 56Fe particles. The number adenocarcinomas are significantly increased which display highly aggressive and invasive characteristics of malignancy, which more closely mimics clinical features of lung cancer in humans. Whole genome mRNA analysis was performed and unsupervised cluster analysis of resultant gene expression profiles has suggested divergent responses between mutant and wild type littermate lung associated with radiation type.

Specific Aim 2. Generate or obtain other transgenic mice susceptible to lung cancer to determine the effects of low- and high-LET radiation in both living animals and in isolated tissues. These include the mTERT knockout mouse and the EGFR TK domain mutation mouse. We will initiate pilot experiments of the mTERT KO mouse during the next review period to determine if more end joining, end-end fusions, cytogenetic alterations, micronuclei etc occur when these mice are exposed to 1 and 3 Gy of X-ray. The EGFR mouse model we have obtained has a EGFR mutation that can be induced by tetracycline in the food or drinking water.

Core B. Cell Culture

We placed HBECs into matrigel overlying a lung fibroblast (IMR90) feeder layer and observed that the cells could efficiently differentiate into spheroids or cyst like structures that express SP-A and morphologically resembled small bronchioles and alveolar sacs, and also had lamellar bodies which is a hallmark of type II alveolar cells.

Core C. Genomics and Proteomics Core

This core has started to develop reverse Phase Protein Lysate Arrays (RPPAs). This will allow us to examine the expression of one to several proteins in a moderate throughput platform (hundreds of samples simultaneously). We have also perfected the use of comparative genomic hybridization arrays (aCGH). This will allow us to examine chromosome copy variation which is likely to be important for the analysis of cancer progression studies being carried out in Project 3.

Core D: Biostatistics and Bioinformatics

A centralized database with a server-client architecture has been created to store and share data generated with the various NSCOR experiments. This multi-user database system has already been implemented and extensively tested in other projects and is being adapted and expanded more specifically for NSCOR. The database allows easy access and sharing of all NSCOR data by all investigators. Thousands of collected samples and large sets of data can be organized and shared across investigators and statisticians. Its centralized nature makes it possible to schedule regular backups, to ensure that data are organized in a way that analysis and data interpretation are facilitated, and to allow investigators to keep track of experiments and samples and thereby substantially minimize errors. The database is accessible through the Internet making it possible to enter data in real-time from remote sites such as BNL.

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